DS 2500 *Fall 2025* Lab Exercise 06

Deadline : *10/20/2025 11:59 PM EST*

Late Deadline: *11/03/2025 11:59 PM EST with 50 points off*

Total Points: *200 pts*

Grading:

- This lab assignmnet is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (`.py`) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Additional Instructions:

- You are welcome to use external packages to test whether your solution works. However, you may not use any external packages within your function implementations.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Problem 1 (50pts)

- Write a function called `find_average()` that takes a list of numbers and returns the mean of the list.
  - If an empty list is passed, the funtion should return `None`.

Problem 2 (50 pts)

- Write a function called `find_median()` that takes a list of numbers and returns the median of the list.
  - If an empty list is passed, the funtion should return `None`.

Problem 3 (50 pts)

- Write a function called `find_mode()` that takes a list of numbers and returns the mode of the list.
  - If there are multiple modes, the function should return all of them in a list.
  - If an empty list is passed, the funtion should return `None`.

Problem 4 (50 pts)

- Write a function called `find_variance()` that takes a list of numbers and returns the population variance of the list.
    - If an empty list is passed, the funtion should return `None`.

Extra Problem 1 (0 pt; not graded)

```
Dataset EP1

ID  Hours Studied   Score   Result
1       10          85      Pass
2       18          90      Pass
…       …           …       …
98      10          50      Fail
99      15          70      Fail
100     15          88      Pass
```

- What types of visualizations can be created based on Dataset EP1?